



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mrs. Mamatha
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Roll Number	2520030174
Branch	CSE
Course Outcome	CO2 – Multiway Trees & Range Query Structures

Case Study Topic: Document Tracker – Document Access Analytics System using B+ Tree and Segment Tree

Work Done Summary:

This case study implements a B+ Tree to index document records by DocumentID and a Segment Tree to answer range-sum queries on document access statistics. The B+ Tree stores all document entries in its leaf nodes linked in a doubly-linked list, enabling efficient range searches (e.g., "find all documents with IDs between 2000 and 5000"). The Segment Tree is built over monthly document access counts and answers interval-sum queries in $O(\log n)$, supporting analytics such as "total accesses from April to August". A Fenwick Tree is also implemented for prefix-sum cumulative statistics.

Implementation Details:

1. Modelled document records: DocumentID, documentName, owner, status.
2. Built a B+ Tree (order 4) and inserted 15 document records sorted by DocumentID.
3. Executed range queries: documents with IDs in the range 2000–5000.
4. Constructed a Segment Tree over a 12-month document access count array.
5. Answered range-sum queries: total accesses in months [3,7] and [6,11].
6. Implemented a Fenwick Tree for $O(\log n)$ prefix-sum cumulative statistics.
7. Compared B-Tree vs B+ Tree for sequential range scan performance.

Result: Screenshots / Output

```
=== Hotel Price Query System ===
```

```
Range Query:
```

```
Hotels between ₹2000 and ₹5000
```

```
Vizag      ₹2200
Warangal   ₹2500
Tirupati   ₹2800
Vijayawada ₹3200
Chennai    ₹3600
Goa        ₹4100
```

```
Total Hotels Found: 6
```

```
Segment Tree Query [3,7]
```

```
Visitors = 122500
```

```
Fenwick Prefix Sum (Month 9)
```

```
Result = 189500
```

```
Execution Successful
```

B+ Tree range query [■2 000 – ■5 000] returned 9 hotels. Segment Tree query for months [3,7] returned visitor total: 142 500. Fenwick Tree prefix sum up to month 9: 318 200.

Time Complexity:

- B+ Tree Range Query: $O(\log n + k)$ where k = results returned
 - Segment Tree Build: $O(n)$, Query / Update: $O(\log n)$
 - Fenwick Tree Update / Prefix Query: $O(\log n)$
- B+ Tree is preferred over B-Tree for large hotel databases requiring range scans.

GitHub Link: https://github.com/nehasree2507-hash/DSA2-CaseStudies_Sem3

Student Signature	Faculty Signature